ARPads: Mid-air Indirect Input for Augmented Reality

ABSTRACT

We investigate the concept of indirect input for interacting with virtual objects displayed in an Augmented Reality headset. Similarly to how a trackpad decouples cursor movements from finger movements, an *ARPad* allows users to control the cursor position through movements performed in a mid-air plane, which is not colocated with that of the screen. We introduce a design space for designing different *ARPads*, taking into account the position of the pad relative to the user's body and the orientation of the pad relative to that of the screen. We report on a study showing that 1) indirect input competes with direct gaze or hand input in terms of performance, while being less fatiguing, that 2) an *ARPad* is more efficient when it is attached to users' wrist or waist than when attached to their thigh, and that 3) the *ARPad* and the screen should have the same orientation.

Author Keywords

AR headset; Indirect input; Small scale gestures

CCS Concepts

•Human-centered computing → Mixed / augmented reality; Gestural input;

INTRODUCTION

Augmented Reality (AR) headsets make it possible for users to get private, in-situ virtual representations. They can offer personalized exhibits in museums (*e.g.*, [37]), deliver specific instructions in an educational context (*e.g.*, [16]), or display any data visualization related to objects in the environment [43]. Interacting with such representations is typically achieved through gaze and overhead hand gestures. For example, the Microsoft Hololens requires users to move their head to move the cursor over the object of interest, and then raise their arm to perform an air-tap hand gesture in the trackable area that is ahead of the headset.

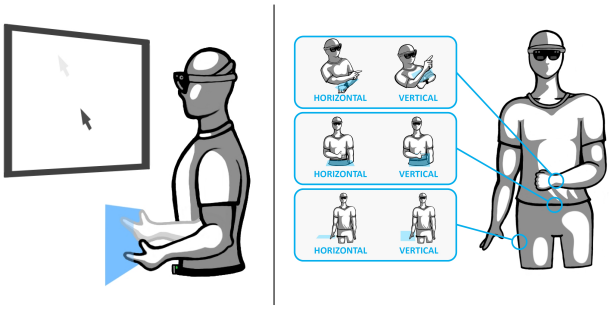


Figure 1. (Left) Mid-air indirect input: hand movements within an *ARPad* control the location of the cursor in the headset screen. (Right) The six different *ARPads* that we evaluate. *ARPads* are mid-air virtual planes that are invisible. They are represented in translucent blue here for illustration purpose.

Both gaze input and overhead gestures have drawbacks. Gestures performed with arms in an upward position are tiring [21, 4], and gaze-based input is *unnatural as it overloads a perceptual channel [...] with a motor control task* [50]. In the case of AR headsets that estimate gaze based on head position and orientation, the problem is even worse as moving the head can make peripheral (but potentially interesting) objects leave users' field of view. Furthermore, performing unnatural head movements can be socially awkward as performing overhead gestures can be [1]. This results in a mismatch between motor actions that seem exaggerated to interact with visual representations that are intended to be for private, potentially confidential use.

We investigate the concept of *ARPad* as a means to replace or complement input with AR headsets. An *ARPad* can be seen as a virtual plan within which users perform small scale movements to control the cursor displayed in the headset's screen (Figure 1-Left), like a trackpad does with a cursor displayed on a laptop's screen.[1] As already demonstrated in the context of remote control of distant vertical screen [29], this indirect way of controlling the cursor results in more relaxed postures and more compact gestures that are both comfortable to perform and acceptable in public [12].

[1]The parallel is not to be considered literally as a trackpad does relative cursor control (with clutching actions) while our implementation of an *ARPad* does absolute control (as a graphics tablet does).

We contribute a *design space* for such indirect input in the context of AR headsets. As illustrated in Figure 1-Right, we consider two dimensions: where the gestures are performed relative to the user's body (wrist, waist or thigh), and what the main orientation of those gestures is (vertical, horizontal). We conduct a study that evaluates points in this space as well as two direct input baselines (gaze input, and overhead input) on both discrete and continuous input tasks. Our study supports the following messages: 1) indirect input competes with direct input in terms of performance, while being less fatiguing than overhead gestures; 2) an *ARPad* is more efficient when positioned close to users' wrist or waist than to their thigh; and 3) the *ARPad* and the headset screen should have the same orientation.

RELATED WORK

Indirect input is a fundamental concept in HCI that received a lot of attention in the literature about pointing. We focus on works that are tightly related to our contribution with a review of what has been done regarding interaction with head-mounted displays with a focus on mid-air gestures and how they have been used for indirect input.

Interacting with Head-mounted Displays

Usability is a key aspect to consider if we want to facilitate adoption of AR glasses [26]. Many research efforts have been recently done to improve interaction with such devices. As mentioned above, interaction with current AR headsets usually relies on gaze input. While it is fast, gaze input also suffers from precision issues [48]. Gaze input thus benefits from being complemented with other modalities [27] such as a mouse in a desktop setting [50] or a handheld touch device in a mobile setting [45]. Gaze input also suffers from the Midas Touch problem [22, 23], which can cause accidental activation commands if no proper selection mechanism is implemented. Selection with gaze is usually performed using a dwell action, or it can rely on an additional device [48, 22]. In the specific case of moving targets, selecting a target can also be achieved through eye pursuits [24, 40].

Mid-air gestures have also received a lot of attention for interacting with head-mounted displays. They avoid relying on additional devices and allow users to perform *direct manipulation* of objects that are within arm's reach [31]. In the specific context of AR headsets, gaze and mid-air input can also complement each other. Commercial solutions like the Microsoft Hololens relies on air tap gestures for selection, and some research prototypes propose larger sets of gestures (*e.g.*, Gaze+Pinch gestures in [39]). Optical sensors for tracking hand gestures are usually mounted on the headset, looking in the direction of users' sight (*e.g.*, [8]). This means that users have to perform gestures at a relatively high height, which quickly causes discomfort and fatigue [21].

When users do not need to keep their hands free, they can also rely on portable multi-touch devices for interacting with head-mounted displays [46]. They can even rely on a desktop input device such as a mouse when their tasks can be performed in the office environment [16]. However, portability and empty hands are often desirable with wearable displays. A hands-free alternative consists in using users' body as an input device with large amplitude gestures performed on the whole body [14] or with finger gestures on specific body parts (*e.g.*, on the forearm [20], on the hand palm [34], on the abdomen [47], or even on the face [28, 42, 49]). Wearable devices also support portability. Dobbelstein *et al.* have proposed wearables for interacting with headset screens with *e.g.*, a touch-enabled belt [9] and pocket [10]. Finally, foot taps have also been recently proposed as a means to perform menu selection with head-mounted displays [35]. However, all those solutions raise different questions related to social acceptance, to sensor availability, or to portability of the equipment which is necessary for tracking.

Overcoming Fatigue of Mid-air Gestures

In the light of the above, mid-air gestures remain a very good candidate for interacting with headset screens. However, how to track such gestures in a portable setting is not trivial. Some research prototypes embed cameras in clothes, shoes or jewels. For example, the Gesture Pendant [15] is a wearable device used to control home environments with mid-air gestures. SixthSense [32] extends the Gesture Pendant by combining a projector and a camera in a pendant. It can project digital content in the physical environment in front of the user, and capture gestures that are performed at chest height. ShoeSense [5] augments a shoe with a Leap Motion to track large amplitude arm and hand gestures to control wearable applications. All these wearable tracking equipments have the advantage of allowing users to perform mid-air gestures with arms in a lower position than they are when gestures are tracked with an optical sensor mounted on the headset itself.

OwnerShift [13] is a recent technique for Virtual Reality (VR) headsets that makes users able to progressively relax their posture during long lasting interactions. With OwnerShift, users see a virtual representation of their hands, which is progressively shifted up so as to encourage users to shift their real hands down, smoothly transitioning from overhead to waist. OwnerShift builds upon the set of techniques that play with the rubber hand illusion [6]. For example, the Go-go Interaction Technique virtually extends users' arms to reach distant objects without walking [41]. The complementary approach also works: instead of virtually extending users' body, virtual objects can be brought closer in the motor space while preserving their positions in the visual space [33]. However, AR gives less freedom than VR as it does not make it possible to modify users' body representation.

Gunslinger [29] addresses the fatigue problem of mid-air gestures in a more radical manner. Relying on what has been later called *at-your-side gestures* [44] and earlier used for zooming actions in [36], Gunslinger allows users to do remote control of a large display using small-scale movements that are performed in a relaxed posture. One Leap Motion controller is attached to each thigh, tracking both hands when arms are along the body. Users point with their dominant hand following a trackpad metaphor, and activate commands with postures performed with their non-dominant hand. Gunslinger support for pointing is very similar to a horizontal *ARPad* attached to users' thigh in our design space, with a difference in imple-

mentation though: Gunslinger works in relative mode (with clutching actions) while our implementation of *ARPads* work in absolute mode.

Extending Input Space in the Air

Extending the interaction space in the air is a strategy that has been implemented for other devices than head-mounted ones. It has initially received a lot of attention with small devices because of the limited size of their interaction surface, which causes occlusion and precision issues. The Gesture Watch [25] allows users to perform mid-air gestures above the watch. Gestures are performed with the dominant hand over the watch worn on the non-dominant arm (the watch is equipped with an array of proximity sensors). The Abracadabra wristwatch [19] extends a watch's sensing area above the surface in the same way, but differs in the sensing technology: the finger from the dominant hand wears a magnet, making it trackable by the watch's magnetometers. Here again, those techniques share some similarities with *ARPads* when they are located at users' wrist with a horizontal orientation.

Building upon the concept of *continuous interaction space* that considers a touch surface and the space around it as a continuum [30], Chen *et al.* propose a smartphone that is equipped with a depth camera in order to extend the vocabulary of possible commands with air+touch gestures [7]. The combination of touch and mid-air input have also proven efficient for performing large amplitude slide gestures on a smartphone [3] or on a smartwatch [18].

Our contribution with respect to related work:
Our related work shows that the concept of mid-air indirect input is not new, but it has been implemented in an ad hoc manner here and there. Our contribution is a unifying one, with an organization of such techniques in a *design space* and a *comparative evaluation* to assess their advantages and limits in order to deliver *take-away messages for practitioners*.

ARPADS DESIGN SPACE

CA: Imaginary interfaces as introduced by Gustafson *et al.* [17] are interfaces that can be operated with empty hands, and without any visual support. While imaginary interfaces were envisioned with direct input, *ARPads* can be thought of as imaginary input devices to control a cursor on a non-colocated screen.

Design Principles

The initial scenario that leaded to the project is a simple collaborative task using a wall-size display and AR-headsets. This kind of setup can be useful while interacting with both private and public data at the same time. Users have access to the public data via the wall-size display and to the private's ones via the headsets. The idea was for the user to be able to interact with both the two devices easily even if the input techniques are different. Keeping the hands of the user free was the best way to make this double interaction easy as the wall-size display usualy asks for touch input. That is the reason why we decided to remove any tangible controller from this scenario. This leaded to the first guideline of the project which is the concept of hand-freedom.

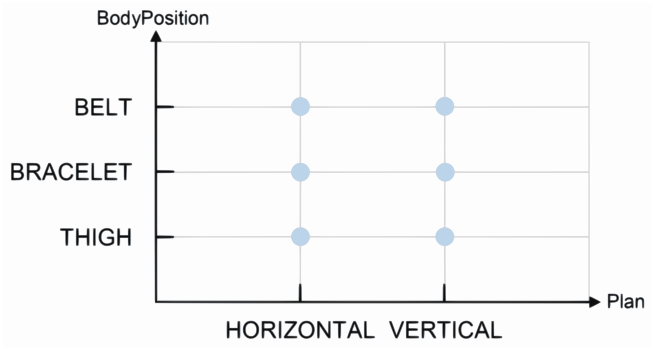


Figure 2. The two dimensions of the design space represent a total of six different possible positions for the mid-air trackpad.

G_1: The hands of the user should remain free of any controller.

Once the hands of the user are free, the question of how to interact with the headsets is the next problem to solve. If we put aside all the tangible controllers, gestures is the current input used for both VR and AR headsets. Unfortunately, these gestures have to be visible from the headset, which means that users should interact in front of their head. This kind of behaviour can be percieved as unusual by the social environement of the user. This is the reason why we would like to avoid such large gestures in public areas.

G_2: Gestures should not be too much perceptible by the social environment of the user.

Finally, interacting at the head level can also be very tiring for users as they need to hold their arm up. This statement is especially true when the user has to perform continuous pointing actions. Consequently, the last objective of this design space will be to prevent the user from experiencing too much tiredness while interacting with the system.

G_3: The gestures performed should have a low physical demand.

Design

To match G_1, we had the idea of creating a "mid-air trackpad" to manipulate a 2D cursor in AR. This trackpad woud be stuck to users, so that they could interact with it at any time without anymore effort. The concept of "mid-air trackpad" is inspired from the Imaginary Interfaces of Gustafson *et al.* [17]. In Imaginary Interfaces, a camera is sticked to the user's chest and operates as a direct input without any visual feedback. What we would like to do is to extend this concept to an indirect input device that could control an AR cursor in a non-colocated screen. We would also like to test if the location of such an interface could affect either the physical demand or the pointing performances of the users.

The position of the trackpad relatively to the user's body could have a great impact on both the social acceptance and the tiredness of the user while using the system. Consequently, the body position BODYPOSITION will be considered as a dimension in the design of our system. The choosen body positions should help the user to keep a comfortable posture while manipulating the 2D cursor.

We identified three different values for this dimension that could be convenient to use for users. These values represent the possible position of the trackpad on the user's body:

- Belt: Using a trackpad situated on the belt would allow users to interact in front of their body, which could be convenient.

- Bracelet: If the trackpad is located on the wrist, the user is able to adjust the trackpad by simply moving the hand such as smartwatches. This add an interesting degree of freedom.

- Thigh: As presented in Gunslinger [29], interacting close to the thigh allows to keep the arm in a comfortable position.

Our trackpad has only two dimensions, meaning that all the mid-air movements of the user should be projected on a plan. This could be projected on any plan in the 3D space. To make the design simple we decided to keep only two values:

- Vertical

- Horizontal

These two dimensions combined represent a total of six different mid-air trackpads (see **??**).

ARPADS IMPLEMENTATION

CA: Like OwnerShift [13] and Gunslinger [29], ARPads rely on Leap Motion controllers for tracking users' hands. With OwnerShift, there are two LMs mounted on the headset: one looking in front of the head, and one looking in front of the waist. With Gunslinger, the user wears two LMs, one attached to each thigh.

Technology

To simulate the mid-air trackpad, and be able to match the theroritical design of *ARPad*, we needed a light-weight device that could detect the hands' movements of a user. Like OwnerShift [13] and Gunslinger [29], *ARPad* relies on Leap Motion controllers for tracking users' hands. With OwnerShift, there are two LMs mounted on the headset: one looking in front of the head, and one looking in front of the waist. With Gunslinger, the user wears two LMs, one attached to each thigh. With *ARPad*, we use only one hand and the plan will remain the same. Consequently, we need only one Leap Motion (Orion 4.0.0+52238) at the time. Unfortunately, the device has some limits: if a hand is too close to the device (< 10cm), the detection will get lower and lower, until it stops working. In a context where the controller is placed on users body, the physical demand of a technique could be increased if users have to maintain their hands at a certain distance from their body. The Leap Motion is hold in place using pieces of velcro tape.

System

We use the Microsoft HoloLens as an AR-Headset. Unfortunatelly, as the Leap Motion can not be directly plug to the headset, we use an external computer linked to the controller to simulate our scenario. The Leap Motion sends data to the computer via a USB cable. The data that contain the position of the user's hand are send to the headset using sockets.

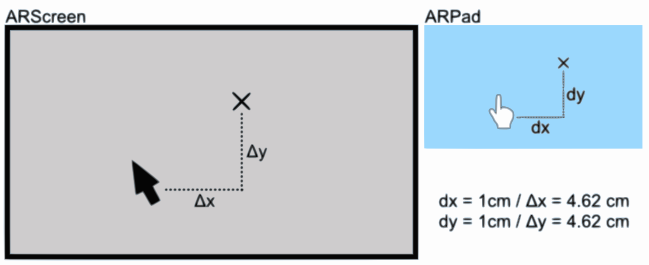


Figure 3. Scale factor between the *ARPad* and the ARscreen.

Implemented Techniques

We implemented few indirect input techniques related to three different body positions. For all of this body positions, the user can interact with the ARscreen using two possible mid-air plans: *Vertical* and *Horizontal*.

BeltHorizontal: Belt LeapMotion using a Horizontal plan;

BeltVertical: Belt LeapMotion using a Vertical plan;

BraceletHorizontal: Bracelet LeapMotion using a Horizontal plan;

BraceletVertical: Bracelet LeapMotion using a Vertical plan;

ThighHorizontal: Thigh LeapMotion using a Horizontal plan;

ThighVertical: Thigh LeapMotion using a Vertical plan.

For each of these techniques, the AR cursor is linked to an ARscreenand not to the headset itself. That means that the head's vibrations have no impact on the position of this cursor. We decided to keep the same scale factor for all the techniques to avoid adding noise between them during the evaluation. We obtain a uniform scale factor of 4.62. This means that moving the hand of 1 cm in front of the *ARPad* moves the cursor of 4.62 cm on the ARscreen(see Figure **??**).

Interaction

ARscreen

Users interact with a horizontal screen of 47.5 x 84.5 cm. This screen is placed at a distance of 194 cm from the user at the begining of a scene and will stay in place until the next one. The cursor is linked to the ARscreen, taking the center of it as the reference for the hand's movements.

ARPad

The Leap Motion has its own coordinate system. All the hand's positions are given in millimeters following an orthonormal coordinate system. For the ease of use, we wanted to have a mid-air plan with the same ratio as the interactible screen that could match the body possible actions of the users that we observed during the pilot studies. We defined that the interactible screen would represent 4.62 times the mid-air plan. Consequently, the *ARPad* would have a size of 10.3 x 18.3 cm. We also adjust the *ARPad* to make it fit a confortable position for the user (defined during the pilot studies) for each of the body positions.

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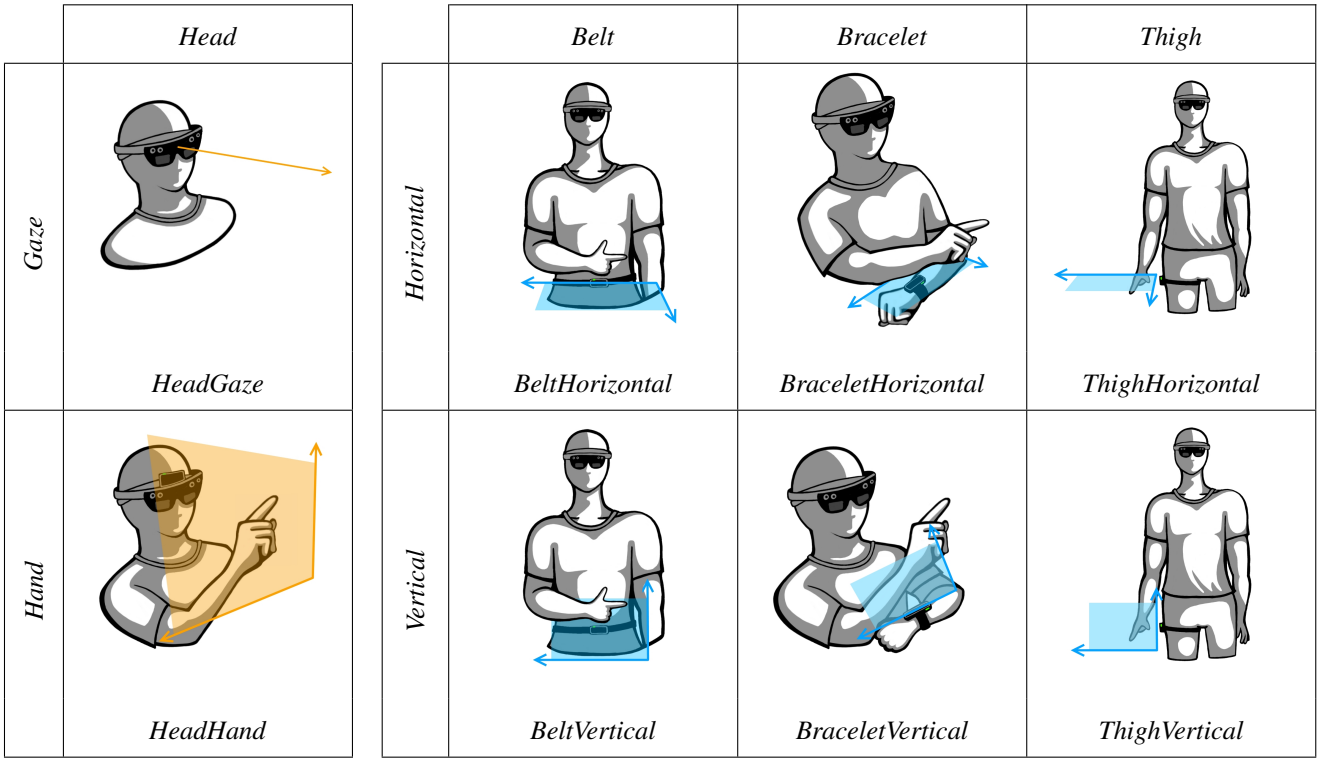


Figure 4. Representation of the eight techniques designed.

EXPERIMENT

CA: Liu *et al.* [29] reports on pointing throughput. Might be good to compare to ours. Note that they point in a relative mode – a posture (thumb toward the palm) is dedicated to clutching.

CA: We should not forget to justify why we use dwell in the study. It is because we focus on users' motor performance and we want to avoid any confounding effect that would come from errors from the gesture recognition engine. Also, we need to make quickly clear that we address 2D selection. It can support interaction with 3D content like a mouse pointer can support 3D object selection on a desktop but the depth dimension in an *ARPad* is ignored. The same (interaction to interaction with 2D content) is done in [27, 35].

Introduction and goals

For the evaluation, the goal was to compare the six indirect input techniques to our baseline which is represented by two direct input techniques. The first one is the *HeadHand*. Using this input technique, users uses their hand as a pointer. In our study, this means that the cursor is visually located at the exact position of the user's hand. The second technique is the technique currently used in the AR-headsets, *HeadGaze*. For this tehnique, the cursor is represented where the head of the user is pointing.

Based on the statement that indirect input is usually more cognitively demanding than direct input for users, we make the hypothesis that *HeadHand* and *HeadGaze* should have the best performances for the two pointing tasks.

H_1: users reach better performances in pointing tasks using direct input techniques.

However, using such direct input techniques can be very tiring. Holding an arm up is definitively physically demanding for a user. The *HeadHand* technique should then be the most tiring technique as it asks for the user to keep the arm up while interacting with the ARscreen. This statement could have an impact on the actual performances, of the users. That is why we are really interested in the measure of this tiredness that users could feel during the two tasks. This leads to our second hypothesis.

H_2: users are less tired while pointing using an indirect input technique.

Another important factor that may impact the measures got from the evaluation is the plan used for the *ARPad*. It can be either *Horizontal* or *Vertical*. For all the indirect techniques, the vertical plan corresponds to an *ARPad* which is colocated with the ARscreen contrary to the horizontal plan which is not. For both *HeadGaze* and *HeadHand*, there is no *ARPad*and the input's plan remains the same as the ARscreen so we also call these plans colocated. It may seem logical that interacting without doing any mapping between the ARscreen and the *ARPad* should be quicker for users. But as we are used to doing these transformations by using input devices such as trackpads, this could also have no impact on our results. Still, we make the hypothesis that this additional mapping could affect the performances of the users.

H_3: users reach better performances in pointing tasks when the input plan is colocated with the. ARscreen.

Finally, our last hypothesis relies on the assertion of Guiard [ref]. This suggests that there is an asymmetric division of labor in human bimanual actions. The two hands of a user cooperate with each other. The formed kinematic chain should give a little advantage to the bimanual techniques: *BraceletVertical* and *BraceletHorizontal*.

H_4: Bimanual techniques tend to be more performant than unimanual techniques.

Unfortunately, with this body position, a bimanual control of the ARscreen is impossible as the LeapMotion is situated on one of the two hands. This is a clear drawback that will not be taken into account as for the all duration of the study, users will have to use only one of their two hands, no matter which technique is tested.

As explained above, the main goal of this study is to compare the ratio between the efficiency in pointing and the tiredness of each of the eight techniques to identify some guidelines for indirect inputs in AR. To reduce the quantity of noise in the data, we decided to focus on a 2D ARscreen controlled by a 2D *ARPad*. This helped us to make a real separation between the plan of interest and does not take into account the ability of the user to interact in 3D. 2D means that only two dimensions are used at the time, but it would be still possible to control a 3D object that is only interactable in 2D.

Design

Each of the eight techniques tested in this experiment has two main attributes: BODYPOSITION and PLAN. These are the main factors tested in the study.

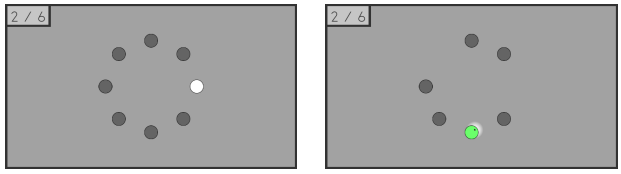
To analyze the impact of these two factors, we run a within-subject experiment with two tasks. One of the task recorded data for discrete pointing actions and the other for continuous pointing actions. As the tiredness of the users while using a technique is what we want to get from the data, we had to put long breaks between techniques to minimize interactions between techniques.

Participants

We had 16 participants (11 men + 5 women) for the evaluation. All of them were right-handed and, except one astrophysicist engineer, all of them worked in the Human-Computer Interaction field. When we asked for any prior experience with AR-headsets, 12 of them answered from never to rarely and 4 of them answered from sometimes to daily. Concerning VR-headsets, 11 of them had a level of expertise from never to rarely and 5 of them from sometimes to daily.

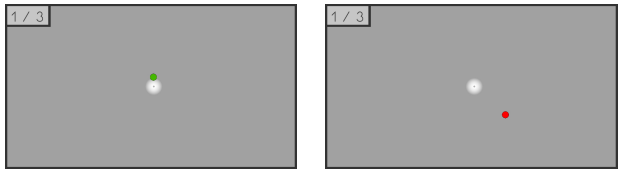
Conditions

The eight techniques implemented for the study are visible in Figure 2. These techniques depend on one BODYPOSITION and one PLAN. Each body position can be linked to two different plans. For *Belt*, *Bracelet* and *Thigh*, the plan can be vertical or horizontal. For the *Head*, the plan represents whether the user is using the hand or the gaze. The direct input techniques are directly linked to the position of the head, which means that, contrary to the six indirect input techniques, they are affected by the vibrations that the head can have.



(a) (b)

Figure 5. Representation of the task 1's screen: a) at the beginning of the task b) while the user is pointing at a target.



(a) (b)

Figure 6. Representation of the task 2's screen: a) while the cursor is close to the target, b) while the cursor is far from the target.

All techniques except *HeadGaze* can also be affected by the hand's vibrations. What we focus on is the motor aspect of each pointing task. Then, we reduced the activity of the user to a simple dowell pointing task in order not to add gesture recognition as noise in the data. Although, unlike Liu *et al.* [29], the mapping we used is direct.

Tasks

Task 1: Discrete, point-based interactions

This task evaluates the ability of a participant to realize discrete interaction with the AR-headset system using one technique or the other. Eight targets are presented to the participant, one of them is white and all the others are grey (see Figure 3). The participant needs to reach the white target that will turn white before disappearing after 0.5 seconds. The targets will turn white in a precise order so that the distance is constant and predictable from the current target to the next one. In this task, we also introduce a second factor which is the width of the target. This should have an impact on the accuracy of the user while reaching the target: the smaller the target is, the more difficult it should be for the user to reach it. This task will be repeated for six trials of eight targets, the three firsts are practices for the participants to get used to the input techniques. The targets' width order is the same for the practice and for the real trials: *Large* (12 meters), *Medium* (6 meters) and *Small* (3 meters).

We have three different measures for this task:

$M_{1,1}$: The completion task: We measure the time that the participant takes to reach a target;

$M_{1,2}$: The error: We record how many times the cursor went in and out of the target before the completion of the task;

$M_{1,3}$: The questionnaire: At the end of the task, we ask the participant to rate the mental demand, the physical demand and the success of the technique used following the NASA-TLX format.

Task 2: Continuous, trajectory-based interactions
The goal of this task was to test the ability of a user to realize a trajectory-based interaction using one technique or the other. This task is inspired by the book of E. Poulton [ref]. It consists of following a predefined path (see Figure **??**) with a cursor using one or the other input technique. A moving target is shown to the user which has to stay as close to it as possible. The task was divided into three trials of 30 seconds each. The three trials corresponded to three different paths A feedback was given to the user depending on its proximity to the target.

For all of the random paths, the targets move through each side of a square of 60x60m. As defined in the book of Poulton [ref], these paths are generated following four sinusoids that depend on the time.

$M_{2,1}$: The mean distance: for every update of the scene, we record the distance between the target and the cursor;

$M_{2,2}$: The standard deviation: We record the global standard deviation of the distance between the target and the cursor;

$M_{2,3}$: The questionnaire.

Apparatus
For this evaluation, we used a Microsoft HoloLens + LeapMotion system. LeapMotion was linked to a Microsoft surface book 2. Communication via wifi using an UDP protocol.

Procedure
The evaluation was run in two sessions of 1H30. For these two sessions, the tasks were grouped by body position and plan (see Figure 5).

At the beginning of the evaluation, participants were asked to sign a consent form and fill a questionnaire about their personal information and their previous experience with AR-headset and VR-headset. Each of the two sessions had the same number of tasks to complete and was conducted the same way. A first BODYPOSITIONwas given to the user following by a first PLAN. First, a physical plan was shown to the user to illustrate where the interaction would take place. Then, there was a tutorial part with a pictogram that represented the current technique. Once the tutorial was started, the user had to reach all the corners of the screen one by one. After having completed the tutorial, the user could either redo it or start the discrete pointing task. A quick explanation of the task followed by the same pictogram as for the tutorial was shown to the user. Whenever the user was ready, we started the task with the first practice trial. After having reached the eight target and between all the trials, the technique's pictogram was shown again and the user had time to rest if it was needed. When the first task was completed, the user could start the continuous pointing task. As for the first task, an explanation followed by a pictogram of the technique used was shown. Once ready, the participant could start the first trial. Once the three trials were completed, the participant could start over the process with the second plan of the BODYPOSITION. When all the tasks for this body position were completed, the participant could rate each of the two plans in terms of mental and physical demand and success. Then a break of 30 minutes was imposed The same process was repeated with the second BODYPOSITION. After the two first BODYPOSITION, the session was over and would start again the next day. The second session was exactly the same as the first one except that at the end, we asked participants to rate globally each of the eight techniques tested. During this process, they could access all their previous answers for them to be consistent.

Results
We first report on the results of the pointing task, then on the pursuit, and finally on the qualitative results.

Pointing Task
Figure 7 shows the pointing time (*PT*) for each TECH by TARGETWIDTH. An ANOVA for the model $PT \sim \text{TECH} \times \text{TARGETWIDTH}$ with the participants as a random factor reveals a significant effects of TECH ($F_{7,105} = 9.02$, $p < 0.001$, $\eta^2_G = 0.16$), and of TARGETWIDTH ($F_{2,30} = 9340$, $p < 0.001$, $\eta^2_G = 0.69$) on *PT*. Moreover, the ANOVA reveals a significant $\text{TECH} \times \text{TARGETWIDTH}$ interaction effect ($F_{14,210} = 3.28$, $p < 0.001$, $\eta^2_G = 0.06$).

TARGETWIDTH has its expected effect on *PT*: *PT* decreases significantly as the target width grow (p'$s < 0.001$). Given the interaction effect we compare the techniques overall, and by target width. This leads to perform $4 \times 28 = 112$ paired t-tests with Bonferroni-Holm corrections applied to the 4 groups of tests ($n = 28$). As the corrections are rather strong (group of 28 p-values), we report both significant ($p < 0.05$) and marginal ($p < 0.1$) results[2] to avoid errors of Type II.

The fastest technique is *HeadGaze*. It is significantly faster than both *Thigh* techniques (p'$s < 0.001$), than *BeltHorizontal* ($p = 0.007$), and than *HeadHand* ($p < 0.001$). For large and medium targets, we have the same results (with similar p's), but, in addition, *HeadGaze* is marginally faster than *BraceletHorizontal* ($p = 0.057$ and $p = 0.088$). However, for the small targets, *HeadGaze* is only significantly faster than *ThighHorizontal* ($p = 0.017$) and *HeadHand* ($p = 0.033$).

The second fastest technique is *BraceletVertical*. It is also significantly faster than both *Thigh* techniques (p'$s = 0.010$), but only marginally faster than *BeltHorizontal* ($p = 0.102$), and than *HeadHand* ($p = 0.095$). The same result holds for medium and large targets, but we have no marginal differences with *HeadHand* (p'$s = 1.0$) and with *BeltHorizontal* in the case of medium targets ($p = 1.0$). For small targets, *BraceletVertical* is, as *HeadGaze*, significantly faster than *ThighHorizontal* ($p = 0.019$) and *HeadHand* ($p = 0.021$).

Then, comes *BeltVertical*, that is only significantly faster than *ThighHorizontal* ($p = 0.013$). For large targets, the same result holds ($p = 0.008$), and in addition *BeltVertical* is marginally faster than *ThighVertical* ($p = 0.058$). However, for small and medium targets we can observe no significant or marginal results.

Finally, *BraceletHorizontal* is neither marginally or significantly faster or slower than any other techniques (but as stated above it is marginally slower than *HeadGaze* for large targets

[2] all the marginal results are significant without correction with all the p'$s < 0.01$.

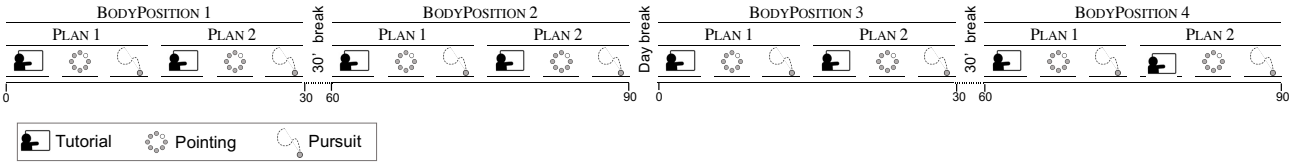


Figure 7. Experiment through time.

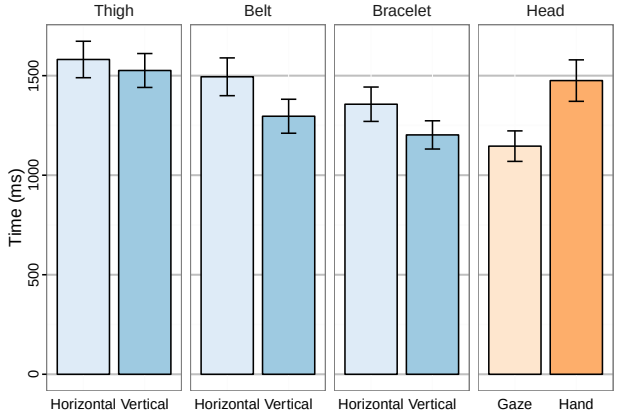


Figure 8. Mean pointing time for all the techniques, grouped by body position and interaction plan.

– the only results concerning targets width). And, both *Thigh* techniques, *BeltHorizontal* and *HeadHand*, the slowest techniques, lead to no other marginal or significant comparisons, overall or by target width.

In summary, ...

Pointing Task: Indirect Techniques

If we focus on the indirect techniques, we can perform an analysis that follow our design space by considering the model: *PT* $\sim$ BodyPosition $\times$ Plan $\times$ TargetWidth. Of course we have a significant effect of TargetWidth ($F_{2,30}=279$, $p<0.001$, $\eta^2_G=0.65$), but we also have significant effects of BodyPosition ($F_{2,30}=12.6$, $p<0.001$, $\eta^2_G=0.10$) and Plan ($F_{1,15}=10.1$, $p=0.003$, $\eta^2_G=0.04$) on *PT*. Moreover, we have only one significant interaction effect: BodyPosition $\times$ TargetWidth ($F_{4,60}=4.02$, $p=0.003$, $\eta^2_G=0.03$).

Thus, participants has been significantly faster with *Vertical* than with *Horizontal*. Moreover, t-tests (corrected $n=3$) show that participant has been: (i) significantly faster with *Bracelet* than with both *Belt* ($p=0.04$) and *Thigh* ($p<0.001$); and (ii) significantly faster with *Belt* than with *Thigh* ($p=0.04$).

Regarding the BodyPosition $\times$ TargetWidth interaction effect, t-tests (correction $n=9$) reveals that: (i) *Bracelet* is significantly faster than *Thigh* for all the targets width, but is significantly faster than *Belt* only for small targets; and (ii) *Belt* is only marginally faster than *Thigh* for large targets (no other marginal/significant differences).

Pointing Task: Errors

The pointing task we design is error-free: the trials end only when the participants succeed to acquire the target. However, we count the number of time the cursor enter and *leave* the target. This number ... OC: should we speak of that? Probably, but we should not go to much in the details

Pursuit Task

For the pursuit task we analyze the average distance to the target (*Dist*) during a pursuit trial.

An ANOVA for the model *Dist* $\sim$ Tech reveals a significant effect of Tech ($F_{7,105}=7.65$, $p<0.001$, $\eta^2_G=0.27$). As suggested by Figure 8, the more efficient techniques are *HeadGaze*, *BraceletVertical*, *HeadHand*, and *BeltVertical*.

Indeed corrected t-tests ($n=28$) reveal that there are no significant or marginal differences between these techniques and that: (i) *HeadGaze* and *BraceletVertical* are significantly more precise than both *Thigh* techniques (all p's<0.003) and *BeltHorizontal* ($p=0.0315$ and $p=0.0154$), and marginaly more precise than *BraceletHorizontal* ($p=0.065$ and $p=0.067$); (ii) *HeadHand* is significantly more precise than both *Thigh* techniques (both p's<0.01), and marginally more precise than *BeltHorizontal* ($p=0.060$); and (iii) *BeltVertical* is marginally more precise than *ThighHorizontal* ($p=0.073$).

If, as we did for the pointing task, we restrict our attention to indirect techniques in the model *Dist* $\sim$ BodyPosition $\times$ Plan, an ANOVA reveals a significant effect of Plan ($F_{1,15}=10.8$, $p=0.005$, $\eta^2_G=0.10$) and a significant BodyPosition $\times$ Plan interaction ($F_{2,30}=6.63$, $p=0.004$, $\eta^2_G=0.09$), but no effect BodyPosition ($F_{2,30}=1.70$, $p=0.200$, $\eta^2_G=0.04$).

Thus, overall, participants has been significantly more efficient with *Vertical* than with *Horizontal*. And, the interaction effect is explained by the fact (t-tests) that this difference is significant for *Belt* ($p=0.033$) and *Bracelet* ($p=0.011$), but not for *Thigh*.

Qualitative Results

Figure 9 shows the results of the answers to the questionnaire.

OC: I recall that we had 2 type of questions, one just after a bodyposition block, that ask à la NASA TLX for mental, perf and fatigue, on a $\sim$20 point scale (!). And an other one that ask to rank the technique for mental, perf and fatigue, at the end of the all expe. Figure 9 shows the NASA TLX normalized on a 5 points scale. We cannot include both in the paper ... I suggest to use the NASA TLX, as this is more standard.

As expected, *HeadHand* is the more tiring technique, and indeed, it has been jugged significantly more tiring than all the other techniques, but *BraceletHorizontal* where the difference is only marginal. We can notice on the graph that the *Bracelet* techniques has been judged more tiring than the other techniques, and that *HeadGaze* is the less tiring technique. In fact,

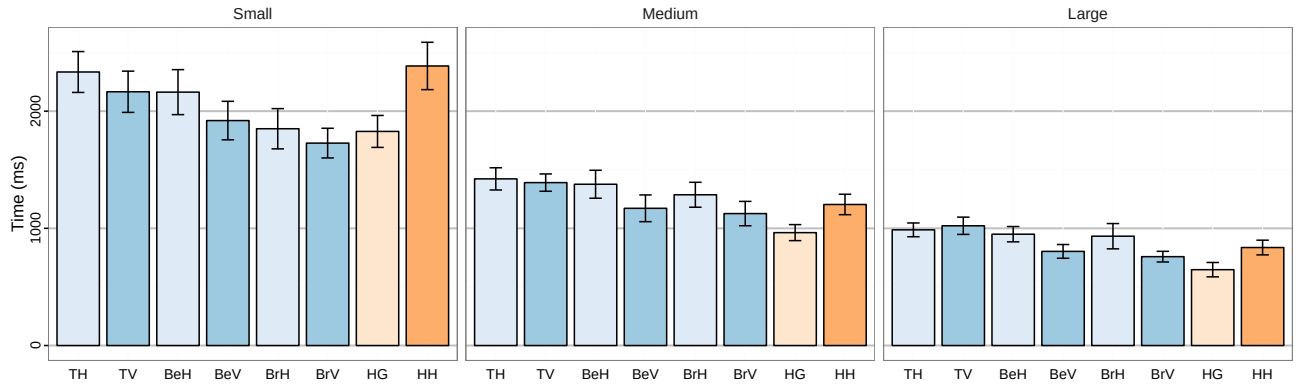


Figure 9. Mean pointing time for all the techniques by target width.

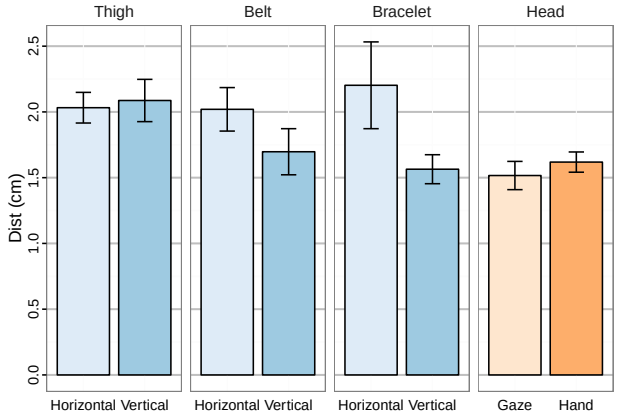


Figure 10. Mean (over the participants) of the average distance to the target during the pursuit, for all the techniques, grouped by body position and interaction plan. OC: what is the unit in the csv? I assumed meter, and transform in cm

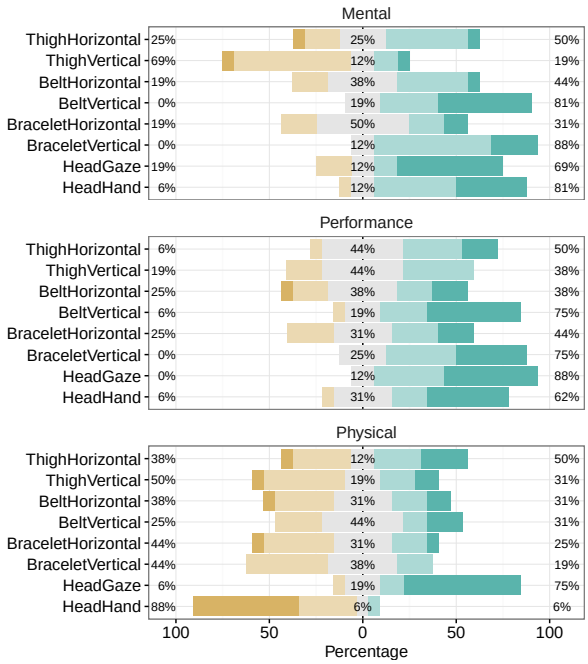


Figure 11. Results of the questionnaire for Mental demand, perceived performance and physical demand. Right (green) is better.

the only other marginal/significant results is that *HeadGaze* is significantly less tiring than *Bracelet* techniques.

Regarding mental demand, we can observe on the graph that *BraceletVertical*, *BeltVertical* and the *Head* techniques are the less demanding techniques, and that the participants had problems with the *ThighVertical* technique. Indeed, the only statistical result is that *ThighVertical* is significantly more demanding than the above mentioned techniques.

Overall Figure 9 suggests that the perceived performance matches the quantitative results: *BraceletVertical*, *BeltVertical* and *HeadGaze* being judged the fastest techniques. From a statistical point of view, the only results are: (i) *BraceletVertical* and *HeadGaze* has been judged significantly faster than *ThighVertical*; and (ii) *BeltVertical* has been judged marginally faster than *BeltHorizontal*;

For each BODYPOSITION we also asked the participants about their preferred interaction PLAN. In the case of *Head*, *all* the participants preferred *Gaze* over *Hand*. Figure 10 shows the result for the indirect techniques. The participants were balanced regarding *Thigh* ($p=0.210$), but we can observe (and

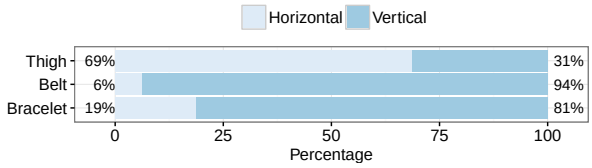


Figure 12. Preferred plan for each body position for the indirect techniques. For Head all the participants preferred Gaze against Hand.

binomial tests show) that participants preferred *Vertical* for both *Belt* ($p<0.001$) and *Bracelet* ($p=0.021$).

Summary

DISCUSSION

CONCLUSION

CA: Would be nice to get some haptic feedback when leaving the pad's area (e.g., users of OwnerShift wear a small vibration motor taped to the index finger) but it requires to instrument users even more.

CA: extend the concept to ARCubes to support 3D input

CA: Note that bracelet is usable while seated for *e.g.*, TV remote control from a couch.

CA: let users define their own ARPAd on the fly

CA: There is no particular reason to make an exclusive choice between direct and indirect input. In the spirit of the Gaze-Shifting technique [38] that switch between direct and indirect input depending on the pen location relative to that of users' sight, input could switch to direct when the user's hand is within the headset's tracking range. We could also imagine bi-manual techniques where one hand performs direct selections, and the other hand performs long, tedious input tasks such as fine parameter adjustments.

CA: Depending on their tasks, their social contexts and their personality traits, users might want their actions with personal devices be transparent to others [11], or almost invisible to others [2]. Subtlety in interaction might be desirable.

ACKNOWLEDGMENTS

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